Random Process

$X(t,\xi)$ an "family" of fuctions.

First-order distribution $F_{X(t)}(x) = P[X(t,\xi) \le x]$

First-order density. $f_{X(t)}(x) = \frac{d}{dx} F_{X(t)}(x)$

The second order distribution of $X(t,\xi)$ is $F_{X(t_1)X(t_2)}(x_1,x_2) = P[\{X(t_1) \le x_1\} \cap \{X(t_2) \le x_2\}]$

- This is sometimes written as $F_X(x_1,x_2;t_1,t_2)$

Similarly, the n-th order distribution is $F_{X(t_1)X(t_2)\dots X(t_n)}(x_1,x_2,\dots,x_n) = P[\cap_{i=1}^{n}\{X(t_i) \le x_i\}]$

- This is sometimes written as $F_X(x_1,x_2,\dots,x_n;t_1,t_2,\dots,t_n)$

The second order density of $X(t,\xi)$ is

$$f_{X(t_1)X(t_2)}(x_1,x_2) = \frac{d^2}{dx_1 dx_2} F_{X(t_1)X(t_2)}(x_1,x_2)$$

- This is sometimes written as $f_X(x_1,x_2;t_1,t_2)$

Similarly, the n-th order density is

$$f_{X(t_1)X(t_2)\dots X(t_n)}(x_1,x_2,\dots,x_n) = \frac{d^n}{dx_1 dx_2 \dots dx_n} F_{X(t_1)X(t_2)\dots X(t_n)}(x_1,x_2,\dots,x_n)$$

For discrete valued processes, we typically specify the joint pmf:

$$p_{X(t_1)X(t_2)\dots X(t_n)}(k_1,k_2,\dots,k_n) = P\left[\bigcap_{i=1}^{n}\{X(t_i) = k_i\}\right]$$

i.i.d process

$\Rightarrow F_{X_1 \dots X_k}(x_1 \dots x_k) = \prod_{i=1}^{k} F(x_i)$

e.g: Bernoulli Random Process $\Rightarrow$ i.i.d process

Sum Processes & i.s.i Processes

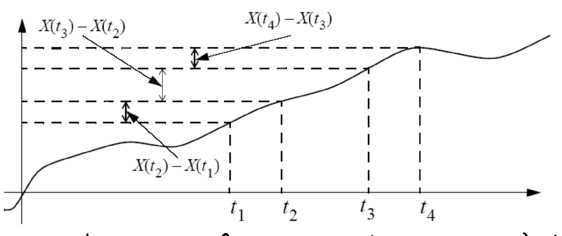
Sum random process

$S_n = \sum_{i=1}^{n} X_i \Rightarrow p_{S_n}(k) = \binom{n}{k} p^k (1-p)^{n-k}$

i.s.i random process

"independent Stationary increment"

Definition: An increment of a random process is the difference between the values of the random process at two different points in time.



independent: if no overlapping $X(t_2)-X(t_1)$ and $X(t_4)-X(t_3)$ is independent

stationary: if $t_4-t_3 = t_2-t_1$ then $X(t_2)-X(t_1)$ and $X(t_4)-X(t_3)$ has same distribution.

e.g. Sum Process $\Rightarrow$ i.s.i process

properties of i.s.i process

$P[\{S_m=j\} \cap \{S_n=k\}] = P[S_m=j \cap S_n-S_m=k-j]$

$= P[S_m=j] \cdot P[S_n-S_m=k-j]$ "independent"

$= P[S_m=j] \cdot P[S_{n-m}=k-j]$ "Stationary" $\Rightarrow S_n-S_m = S_{n-m}-S_0$

Any higher order density can be obtained by using the manipulations on the previous page.
For example, for an integer valued discrete time process,

- For any $n_1 < n_2 < n_3$,

$$P[S_{n_1}=k_1, S_{n_2}=k_2, S_{n_3}=k_3] = P[S_{n_1}=k_1] \times P[S_{n_2-n_1}=k_2-k_1] \times P[S_{n_3-n_2}=k_3-k_2]$$

$$p_{S_{n_1}S_{n_2}S_{n_3}}(k_1,k_2,k_3) = p_{S_{n_1}}(k_1) \times p_{S_{n_2-n_1}}(k_2-k_1) \times p_{S_{n_3-n_2}}(k_3-k_2)$$

- For any $n_1 < n_2 < \dots < n_m$,

$$P[S_{n_1}=k_1, S_{n_2}=k_2, \cdots, S_{n_m}=k_m] = P[S_{n_1}=k_1] \times \prod_{i=2}^{m} P[S_{n_i-n_{i-1}}=k_i-k_{i-1}]$$

$$p_{S_{n_1}S_{n_2}\cdots S_{n_m}}(k_1,k_2,\cdots,k_m) = p_{S_{n_1}}(k_1) \times \prod_{i=2}^{m} p_{S_{n_i-n_{i-1}}}(k_i-k_{i-1})$$

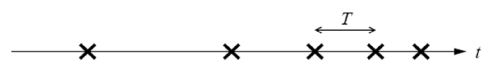
Poisson Process

The Poisson counting process $N(t)$ is the continuous time non-negative integer valued i.s.i. process whose first order density is Poisson:

$$P[N(t)=k] = \frac{(\lambda t)^k}{k!} e^{-\lambda t} \quad \text{for } k = 0,1,2,\dots$$

Property of Poisson Process

The time interval T between adjacent events (the inter-arrival time) is exponentially distributed with parameter λ.



Proof:

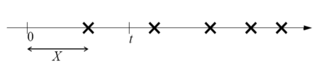
$F_T(t) = P[T \le t]$
$= P[\text{at least one arrival in an interval of length } t]$
$= P[N(t) > 0] = 1 - P[N(t)=0] = 1 - e^{-\lambda t}$

Thus, the cumulative distribution function of T is the same as that of an exponential RV.

- If exactly one event occurs in $[0,t]$, the time it occurs X is uniformly distributed on $[0,t]$.

Proof: Assume $0 \le x \le t$,

$$F_X(x) = P[N(x)=1|N(t)=1] = \frac{P[N(x)=1 \cap N(t)=1]}{P[N(t)=1]}$$
$$= \frac{P[N(x)=1]P[N(t)-N(x)=0]}{P[N(t)=1]} = \frac{P[N(x)=1]P[N(t-x)=0]}{P[N(t)=1]}$$
$$= \frac{\lambda x e^{-\lambda x} e^{-\lambda(t-x)}}{\lambda t e^{-\lambda t}} = \frac{x}{t}$$



Mean & AutoCorrelation of Random Process

① $E(X)$, $Var(X)$

i.i.d: $\begin{cases} E(X_N) = \mu \rightarrow \text{constant.} \\ Var(X_N) = \sigma^2 \rightarrow \text{constant.} \end{cases}$

i.s.i $\begin{cases} E(X_N) = n\mu \rightarrow \text{linear} \\ Var(X_N) = n\sigma^2 \rightarrow \text{linear} \end{cases}$

② Autocorrelation, Covariance

Autocorrelation : $R_X(t_1,t_2) = E[X(t_1)X(t_2)] = \int\int xy f_{X(t_1),X(t_2)}(x,y)dx$

Covariance :

$$C_X(t_1,t_2) = \text{Cov}(X(t_1),X(t_2))$$
$$= E[(X(t_1)-m_X(t_1))(X(t_2)-m_X(t_2))]$$
$$= R_X(t_1,t_2) - m_X(t_1)m_X(t_2)$$

$\text{Var}[X(t)] = C_X(t,t)$

Correlation coefficient : $\rho_X(t_1,t_2) = \frac{C_X(t_1,t_2)}{\sqrt{C_X(t_1,t_1)}\sqrt{C_X(t_2,t_2)}}$

i.i.d. process

$C_X(m,n) = \begin{cases} \sigma^2 & m=n \\ 0. & m \ne n. \end{cases}$

$R_X(m,n) = C_X(m,n) + \mu^2$

i.s.i process

- For an independent stationary increment (i.s.i.) process, $C_S(m,n) = \sigma^2 \min(m,n)$

Proof

Assume first that $m \ge n$, then

$$C_S(m,n) = E[S_m S_n] - E[S_m] \cdot E[S_n]$$
$$= E[(S_n + S_m - S_n)S_n] - E[S_m] \cdot E[S_n]$$
$$= E[S_n^2 + (S_m - S_n)S_n] - E[S_m] \cdot E[S_n]$$
$$= E[S_n^2] + E[S_m - S_n] \cdot E[S_n] - E[S_m] \cdot E[S_n]$$
$$= E[S_n^2] + E[S_m]E[S_n] - E[S_n] \cdot E[S_n] - E[S_m] \cdot E[S_n]$$
$$= E[S_n^2] - E[S_n] \cdot E[S_n]$$
$$= \text{VAR}[S_n] = \sigma^2 n$$

independent increments

Gaussian Random Process

Gaussian Random Process

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- **Definition:** A random process is said to be Gaussian if all finite order distributions are jointly Gaussian distributed, i.e., for any $k < \infty$ and any set of k sample times $t_1, t_2, \dots, t_k$,

$$f_{X(t_1)X(t_2)\dots X(t_k)}(x_1,\dots,x_k) = \frac{1}{(2\pi)^{\frac{k}{2}}|C|^{\frac{1}{2}}} e^{-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{m})^T C^{-1}(\boldsymbol{x}-\boldsymbol{m})}$$

where $\boldsymbol{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_k \end{bmatrix}$, $\boldsymbol{m} = \begin{bmatrix} E[X(t_1)] \\ \vdots \\ E[X(t_k)] \end{bmatrix}$ and $C = \begin{bmatrix} C_X(t_1,t_1) & \cdots & C_X(t_1,t_k) \\ \vdots & \ddots & \vdots \\ C_X(t_k,t_1) & \cdots & C_X(t_k,t_k) \end{bmatrix}$

Stationary Random Processes

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Definition: A process is *stationary* if the joint distribution of any set of samples does not depend on the placement of the time origin, i.e.,

$$F_{X(t_1),X(t_2),\dots,X(t_k)}(x_1,x_2,\dots,x_k) = F_{X(t_1+\Delta),X(t_2+\Delta),\dots,X(t_k+\Delta)}(x_1,x_2,\dots,x_k)$$

for all $x_1,x_2,\dots,x_k,t_1,t_2,\dots,t_k$ and Δ.